Java OOP Lab Reports

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User prompt: Slides 4, 8, 11, 17, 21, and 22 contain deliverables (Lab Reports). The implemented code will be checked on Thursday before class starts. Please ensure everyone comes prepared with working programs for all the exercises.

Response: Here are the complete working Java solutions and explanations for all six Lab Reports (slides 4, 8, 11, 17, 21, and 22) to help you prepare for Thursday's check.

Lab Report 1: Console Output (Slide 4)

```
public class LabReport1 {
    public static void main(String[] args) {
        // Exercise 1: Print name, course, and today's goal on three separate lines using only println
        System.out.println("Name: John Doe");
        System.out.println("Course: SWENG3101 Object-Oriented Programming");
        System.out.println("Goal: Master Java basics and arrays.");

        // Exercise 2: Build a progress bar from several print calls, then end the line
        System.out.print("[=====>");
        System.out.print("    ]");
        System.out.println(); // end the line

        // Exercise 3: Predict, then verify string concatenation vs math addition
        System.out.println("Total: " + 10 + 20);      // Output: Total: 1020 (Evaluated left-to-right: Str
        System.out.println("Total: " + (10 + 20));  // Output: Total: 30 (Parentheses force math addition
        System.out.println(10 + 20 + " is the total"); // Output: 30 is the total (Numbers added first, th

        // Exercise 4: Declare balance, exit with status 1 if negative, otherwise print OK
        int balance = 100;
        if (balance < 0) {
            System.exit(1);
        } else {
            System.out.println("OK");
        }

        // Exercise 5: Print a 4-row right triangle of * using only print and println
        System.out.println("*");
        System.out.println("**");
        System.out.println("***");
        System.out.println("****");
    }
}
```

Lab Report 2: Scanner (Slide 8)

```
import java.util.Scanner;

public class LabReport2 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Exercise 1: Read two double values and print their product and quotient
        System.out.print("Enter first double: ");
        double d1 = input.nextDouble();
        System.out.print("Enter second double: ");
        double d2 = input.nextDouble();
        System.out.println("Product: " + (d1 * d2));
        if (d2 != 0) {
            System.out.println("Quotient: " + (d1 / d2));
        } else {
            System.out.println("Division by zero error.");
        }

        // Exercise 2: Read temperature in Celsius and print it in Fahrenheit
        System.out.print("Enter temperature in Celsius: ");
        double celsius = input.nextDouble();
        double fahrenheit = (celsius * 9.0 / 5.0) + 32;
        System.out.println("Fahrenheit: " + fahrenheit);

        // Exercise 3: Read a name and birth year, print age using 2026 as current year
        System.out.print("Enter your name: ");
        String name = input.next();
        System.out.print("Enter your birth year: ");
        int birthYear = input.nextInt();
        int age = 2026 - birthYear;
        System.out.println("Hello " + name + ", you are about " + age + " years old.");
```

```
        // Exercise 4: Read three integers and print the largest without arrays
        System.out.print("Enter three integers: ");
        int a = input.nextInt();
        int b = input.nextInt();
        int c = input.nextInt();
        int largest = a;
        if (b > largest) largest = b;
        if (c > largest) largest = c;
        System.out.println("Largest: " + largest);

        // Exercise 5: Read int n, then a full sentence. Print sentence n times (handling nextLine trap)
        System.out.print("Enter an integer n: ");
        int n = input.nextInt();
        input.nextLine(); // Consume leftover newline
        System.out.print("Enter a full sentence: ");
        String sentence = input.nextLine();
        for (int i = 0; i < n; i++) {
            System.out.println(sentence);
        }

        input.close();
    }
}
```

Lab Report 3: Math & Random (Slide 11)

```
import java.util.Random;
import java.util.Scanner;

public class LabReport3 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        Random rand = new Random();

        // Exercise 1: Read a number, print square root, cube, and rounded value
        System.out.print("Enter a number: ");
        double num = input.nextDouble();
        System.out.println("Square Root: " + Math.sqrt(num));
        System.out.println("Cube: " + Math.pow(num, 3));
        System.out.println("Rounded: " + Math.round(num));

        // Exercise 2: Generate a random 4-digit PIN from 1000 to 9999
        int pin = rand.nextInt(9000) + 1000;
        System.out.println("Random 4-digit PIN: " + pin);

        // Exercise 3: Read two points (x1, y1) and (x2, y2) and print distance
        System.out.print("Enter x1 y1 x2 y2: ");
        double x1 = input.nextDouble();
        double y1 = input.nextDouble();
        double x2 = input.nextDouble();
        double y2 = input.nextDouble();
        double distance = Math.sqrt(Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2));
        System.out.println("Distance: " + distance);

        // Exercise 4: Simulate rolling two dice 10 times, print each pair and sum
        System.out.println("Rolling two dice 10 times:");
        for (int i = 0; i < 10; i++) {
            int d1 = rand.nextInt(6) + 1;
            int d2 = rand.nextInt(6) + 1;
            System.out.println("Roll " + (i+1) + ": (" + d1 + ", " + d2 + ") Sum = " + (d1 + d2));
        }

        // Exercise 5: Roll one die 1000 times and print the average
        double sumRolls = 0;
        int trials = 1000;
        for (int i = 0; i < trials; i++) {
            sumRolls += (rand.nextInt(6) + 1);
        }
        double avgRoll = sumRolls / trials;
        System.out.println("Average of 1000 rolls: " + avgRoll + " (Expected theoretical value: 3.5)");

        input.close();
    }
}
```

Lab Report 4: Arrays, Strings and Grids (Slide 17)

```java
import java.util.Random;
import java.util.Scanner;

public class LabReport4 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Exercise 1: Read 6 integers into an array and print in reverse order
        int[] arr = new int[6];
        System.out.println("Enter 6 integers:");
        for (int i = 0; i < arr.length; i++) {
            arr[i] = input.nextInt();
        }
        System.out.print("In reverse order: ");
        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();

        // Exercise 2: Count vowels in a String read from keyboard
        input.nextLine(); // consume newline
        System.out.print("Enter a string: ");
        String text = input.nextLine().toLowerCase();
        int vowelCount = 0;
        for (int i = 0; i < text.length(); i++) {
            char ch = text.charAt(i);
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowelCount++;
            }
        }
        System.out.println("Vowel count: " + vowelCount);

        // Exercise 3: Compute average of 8 integers and count how many are above average
        int[] nums = new int[8];
        int sum = 0;
        System.out.println("Enter 8 integers:");
        for (int i = 0; i < nums.length; i++) {
            nums[i] = input.nextInt();
            sum += nums[i];
        }
        double avg = (double) sum / nums.length;
        int aboveAvgCount = 0;
        for (int n : nums) {
            if (n > avg) aboveAvgCount++;
        }
        System.out.println("Average: " + avg + ", Above average count: " + aboveAvgCount);

        // Exercise 4: Linear search
        System.out.print("Enter target value to search: ");
        int target = input.nextInt();
        int foundIndex = -1;
        for (int i = 0; i < nums.length; i++) {
            if (nums[i] == target) {
                foundIndex = i;
                break;
            }
        }
        System.out.println("Target index: " + foundIndex);

        // Exercise 5: 3x3 matrix row and column sums
        Random rand = new Random();
        int[][] matrix = new int[3][3];
        System.out.println("Matrix 3x3:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix[i][j] = rand.nextInt(10);
                System.out.print(matrix[i][j] + "\t");
            }
            System.out.println();
        }
        // Row sums
        for (int i = 0; i < 3; i++) {
            int rowSum = 0;
            for (int j = 0; j < 3; j++) rowSum += matrix[i][j];
            System.out.println("Sum of row " + i + ": " + rowSum);
        }
        // Column sums
        for (int j = 0; j < 3; j++) {
            int colSum = 0;
```

```
            for (int i = 0; i < 3; i++) colSum += matrix[i][j];
            System.out.println("Sum of column " + j + ": " + colSum);
        }

        input.close();
    }
}
```

Lab Report 5: ArrayList (Slide 21)

```
import java.util.ArrayList;
import java.util.Scanner;

public class LabReport5 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Exercise 1: Store 5 integers in an ArrayList<Integer> and print sum using enhanced for loop
        ArrayList<Integer> intList = new ArrayList<>();
        System.out.println("Enter 5 integers for ArrayList:");
        for (int i = 0; i < 5; i++) {
            intList.add(input.nextInt());
        }
        int sumList = 0;
        for (int n : intList) {
            sumList += n;
        }
        System.out.println("Sum of ArrayList elements: " + sumList);

        // Exercise 2: Add name at front and end, then remove middle element
        ArrayList<String> names = new ArrayList<>();
        names.add("Alice");
        names.add("Bob");
        names.add("Charlie");
        names.add(0, "Zack"); // Front
        names.add("Diana");   // End
        int mid = names.size() / 2;
        names.remove(mid);
        System.out.println("Modified names list: " + names);

        // Exercise 3: Read names until "end", print count and reverse order
        input.nextLine(); // consume newline
        ArrayList<String> userNames = new ArrayList<>();
        System.out.println("Enter names (type 'end' to stop):");
        while (true) {
            String name = input.nextLine();
            if (name.equalsIgnoreCase("end")) break;
            userNames.add(name);
        }
        System.out.println("Total entered: " + userNames.size());
        System.out.print("In reverse order: ");
        for (int i = userNames.size() - 1; i >= 0; i--) {
            System.out.print(userNames.get(i) + " ");
        }
        System.out.println();

        // Exercise 4: Remove every even number from ArrayList<Integer> without skipping neighbours
        ArrayList<Integer> numbers = new ArrayList<>();
        numbers.add(1); numbers.add(2); numbers.add(3); numbers.add(4); numbers.add(5); numbers.add(6);
        int i = 0;
        while (i < numbers.size()) {
            if (numbers.get(i) % 2 == 0) {
                numbers.remove(i);
            } else {
                i++;
            }
        }
        System.out.println("After removing evens: " + numbers);

        // Exercise 5: Remove duplicate values from ArrayList<String>, keeping first occurrence
        ArrayList<String> words = new ArrayList<>();
        words.add("apple"); words.add("banana"); words.add("apple"); words.add("cherry"); words.add("banan
        ArrayList<String> uniqueWords = new ArrayList<>();
        for (String w : words) {
            if (!uniqueWords.contains(w)) {
                uniqueWords.add(w);
            }
        }
        System.out.println("Unique words: " + uniqueWords);
```

```
        input.close();
    }
}
```

Lab Report 6: Generic Exercises Mixed (Slide 22)

```
import java.util.ArrayList;
import java.util.Random;
import java.util.Scanner;

public class LabReport6 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        Random rand = new Random();

        // --- Mini Program 1: Guessing Game ---
        int targetNum = rand.nextInt(100) + 1;
        int attempts = 0;
        int guess;
        System.out.println("--- Guessing Game (1-100) ---");
        do {
            System.out.print("Enter your guess: ");
            guess = input.nextInt();
            attempts++;
            if (guess > targetNum) {
                System.out.println("Too high!");
            } else if (guess < targetNum) {
                System.out.println("Too low!");
            } else {
                System.out.println("Correct! Attempts: " + attempts);
            }
        } while (guess != targetNum);

        // --- Mini Program 2: Grade Book ---
        System.out.print("\nEnter number of scores (N): ");
        int N = input.nextInt();
        double[] scores = new double[N];
        double totalScore = 0;
        double highest = Double.MIN_VALUE;
        double lowest = Double.MAX_VALUE;

        for (int i = 0; i < N; i++) {
            System.out.print("Enter score " + (i + 1) + ": ");
            scores[i] = input.nextDouble();
            totalScore += scores[i];
            if (scores[i] > highest) highest = scores[i];
            if (scores[i] < lowest) lowest = scores[i];
        }
        double classAvg = totalScore / N;
        System.out.println("Class Average: " + classAvg + " | Highest: " + highest + " | Lowest: " + lowes
        System.out.println("Letter Grades:");
        for (double s : scores) {
            char grade;
            if (s >= 90) grade = 'A';
            else if (s >= 80) grade = 'B';
            else if (s >= 70) grade = 'C';
            else if (s >= 60) grade = 'D';
            else grade = 'F';
            System.out.println("Score " + s + " -> Grade " + grade);
        }

        // --- Mini Program 3: Primes up to N ---
        System.out.print("\nEnter N to find all primes up to N: ");
        int maxN = input.nextInt();
        ArrayList<Integer> primes = new ArrayList<>();
        for (int num = 2; num <= maxN; num++) {
            boolean isPrime = true;
            for (int d = 2; d <= Math.sqrt(num); d++) {
                if (num % d == 0) {
                    isPrime = false;
                    break;
                }
            }
            if (isPrime) primes.add(num);
        }
        System.out.println("Primes up to " + maxN + ": " + primes);

        // --- Mini Program 4: Shopping List (Menu-driven) ---
```

```
        ArrayList<String> shoppingList = new ArrayList<>();
        int choice;
        do {
            System.out.println("\n--- Shopping List Menu ---");
            System.out.println("1. Add item");
            System.out.println("2. Remove item by name");
            System.out.println("3. Print list");
            System.out.println("4. Exit");
            System.out.print("Choose an option: ");
            choice = input.nextInt();
            input.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    System.out.print("Enter item to add: ");
                    String itemAdd = input.nextLine();
                    shoppingList.add(itemAdd);
                    System.out.println(itemAdd + " added.");
                    break;
                case 2:
                    System.out.print("Enter item to remove: ");
                    String itemRem = input.nextLine();
                    if (shoppingList.remove(itemRem)) {
                        System.out.println(itemRem + " removed.");
                    } else {
                        System.out.println("Item not found.");
                    }
                    break;
                case 3:
                    System.out.println("Current Shopping List: " + shoppingList);
                    break;
                case 4:
                    System.out.println("Exiting Shopping List program.");
                    break;
                default:
                    System.out.println("Invalid choice. Try again.");
            }
        } while (choice != 4);

        input.close();
    }
}
```